RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

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Semester : 3rd Semester

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CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 5:24 PM
State	Finished
Completed on	Saturday, 10 October 2026, 5:28 PM
Time taken	4 mins 36 secs
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

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Finish review

Question 1

Correct Mark 10.00 out of 10.00 Flag question

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*
1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int n,i;
    scanf("%d",&n);

    long long dp[n+1];
    dp[0]=1;
    for(i=1;i<=n;i++){
        dp[i]=dp[i-1];

        if(i>=3){
            dp[i]+=dp[i-3];
        }
    }
    printf("%lld",dp[n]);
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 5:47 PM
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Time taken	7 mins 27 secs
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1

Correct Mark 10.00 out of 10.00 Flag question

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int max_val(int a, int b) {
    if (a > b) {
        return a;
    }
    return b;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    int grid[n][n];
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &grid[i][j]);
        }
    }

    int dp[n][n];

    // Starting cell
    dp[0][0] = grid[0][0];

    // Initialize first row (can only move right)
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + grid[0][j];
    }

    // Initialize first column (can only move down)
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + grid[i][0];
    }

    // Fill the rest of the table
    for (int i = 1; i < n; i++) {
        for (int j = 1; j < n; j++) {
            dp[i][j] = grid[i][j] + max_val(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    printf("%d\n", dp[n - 1][n - 1]);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 5:58 PM
State	Finished
Completed on	Saturday, 10 October 2026, 5:58 PM
Time taken	35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int max_val(int a, int b) {
    if (a > b) {
        return a;
    }
    return b;
}

int main() {
    char s1[1005];
    char s2[1005];

    if (scanf("%s", s1) != 1 || scanf("%s", s2) != 1) {
        return 0;
    }

    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max_val(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    printf("%d\n", dp[m][n]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 6:00 PM
State	Finished
Completed on	Saturday, 10 October 2026, 6:01 PM
Time taken	35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int max_val(int a, int b) {
    if (a > b) {
        return a;
    }
    return b;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int dp[n];
    int max_length = 1;

    for (int i = 0; i < n; i++) {
        dp[i] = 1; // Every element alone is a subsequence of length 1
        for (int j = 0; j < i; j++) {
            if (arr[j] <= arr[i]) { // Non-decreasing condition
                dp[i] = max_val(dp[i], dp[j] + 1);
            }
        }
        max_length = max_val(max_length, dp[i]);
    }

    printf("%d\n", max_length);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review